

Periodic Adverse Drug Experience Report (PADER)

For

BISOPROLOL

(Application number: B-1)

Dev Pharma Client

Period covered by this report: <START_DATE> to <END_DATE>

Date of Approval in US: <TO BE FILLED>

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Introduction

Introduction

This Periodic Adverse Drug Experience Report (PADER) summarizes the worldwide safety experience of Bisoprolol for the reporting interval from 2024-12-27 to 2025-12-26, in accordance with post-marketing safety reporting requirements of the FDA under 21 CFR 314.80.

Bisoprolol is indicated for the management of hypertension and heart failure, contributing to improved cardiovascular outcomes in patients.

The product is approved in the United States under Application Number <TO BE FILLED>. It is available as <TO BE FILLED>.

This report provides a summary of adverse events reported during the reporting interval, including serious unlisted cases (15-day Alert Reports), along with relevant regulatory updates and safety information in accordance with 21 CFR 314.81.

Summary Tabulation of Adverse Events

Summary Tabulation of Adverse Events

Summary Tabulation of Adverse Drug Reactions from Postmarketing Sources

Spontaneous, including regulatory authority and literature

The following table summarizes adverse drug reactions reported for Bisoprolol from postmarketing sources during the reporting interval from 2024-12-27 to 2025-12-26. Adverse reactions are presented by MedDRA System Organ Class and Preferred Term and include both serious and non-serious reports received from spontaneous sources, regulatory authorities, and literature reports. Interval and cumulative case counts are provided where applicable.

System Organ Class	Preferred Term	Serious Interval	Serious Cumulative	Non-Serious Interval	Non-Serious Cumulative	Total
Blood and lymphatic system disorders		110	0	0	0	110
	Anaemia	21	0	0	0	21
	Thrombocytopenia	14	0	0	0	14
	Pancytopenia	12	0	0	0	12
	Neutropenia	8	0	0	0	8
	Eosinophilia	7	0	0	0	7
	Agranulocytosis	5	0	0	0	5
	Febrile bone marrow aplasia	5	0	0	0	5
	Febrile neutropenia	5	0	0	0	5
	Leukopenia	4	0	0	0	4
	Immune thrombocytopenia	2	0	0	0	2
	Coagulopathy	2	0	0	0	2
	Normochromic normocytic anaemia	2	0	0	0	2
	Lymphadenopathy	2	0	0	0	2
	Cytopenia	2	0	0	0	2
	Acquired haemophilia	2	0	0	0	2
	Spontaneous haematoma	2	0	0	0	2
	Thrombocytosis	1	0	0	0	1
	Blood loss anaemia neonatal	1	0	0	0	1
	Lymphadenopathy mediastinal	1	0	0	0	1

	Thrombotic microangiopathy	1	0	0	0	1
	Splenic infarction	1	0	0	0	1
	Antiphospholipid syndrome	1	0	0	0	1
	Spleen ischaemia	1	0	0	0	1
	Red blood cell abnormality	1	0	0	0	1
	Bone marrow failure	1	0	0	0	1
	Deficiency anaemia	1	0	0	0	1
	Lymphadenitis	1	0	0	0	1
	Autoimmune haemolytic anaemia	1	0	0	0	1
	Bicytopenia	1	0	0	0	1
	Coagulation factor deficiency	1	0	0	0	1
	Hypoprothrombinaemia	1	0	0	0	1
...	...	...	...	...	...	...
Total		3645	0	3	0	3648

Source: Spontaneous postmarketing surveillance. Interval = current reporting period only.
Cumulative = 0 (no prior reporting period data loaded). Serious cases: events meeting ICH
E2A seriousness criteria.

Case Presentation

Case Presentation

The following 5 unlabelled Preferred Terms were identified in ≥ 3 distinct serious cases during the reporting interval. Narratives are presented by System Organ Class.

Acute kidney injury

During the reporting interval, 80 cases of acute kidney injury were reported in patients receiving Bisoprolol. The patient population included a diverse demographic with a predominance of older males, with ages ranging from 50 to 85 years. Relevant comorbidities included Clostridium difficile colitis in one patient and falls in another, while frequently reported concomitant medications included antihypertensives and diuretics. The clinical course of acute kidney injury generally presented within a few days to weeks of starting therapy, with severity ranging from mild to moderate impairment. Reported outcomes included favorable recovery in 64% of the cases, while others demonstrated ongoing recovery at the time of reporting. Based on the available information, a causal relationship between Bisoprolol and acute kidney injury cannot be established.

Drug ineffective

During the reporting interval, 53 cases of Drug ineffective were reported in patients receiving Bisoprolol. The patient population included a mix of ages and genders, with cases reported from the United Kingdom, Italy, Portugal, and Spain, encompassing both older adults and younger individuals. Relevant comorbidities included chronic kidney disease, heart failure, and palpitations among some patients, while frequently reported concomitant medications varied across individual cases. The clinical course varied, with some patients experiencing ongoing symptoms of ineffectiveness, while others showed improvement or resolution after multiple therapies, although the exact timeline differed across cases. Reported outcomes included unknown status for a few patients, while others had resolved or were recovering. Based on the available information, the causal relationship between Bisoprolol and the reported event remains indeterminate.

Hypotension

During the reporting interval, 46 cases of Hypotension were reported in patients receiving Bisoprolol. The patient population predominantly involved elderly patients, with ages ranging from 40 to 93 years and a slight male predominance. Common comorbidities included cardiovascular disease and diabetes, with frequently reported concomitant medications being antihypertensives and diuretics. The clinical course exhibited variable temporal patterns; several cases were noted to occur following dose adjustments or initiation of therapy, while others presented acutely. Symptoms reported alongside hypotension included malaise, confusional state, chest pain, and dyspnoea. Reported outcomes varied, with some patients reported as recovering or having resolved the event, while others remained ongoing, particularly in more complex cases involving multiple comorbidities. Based on the available information, the causal relationship between Bisoprolol and the reported event remains indeterminate.

Drug interaction

During the reporting interval, 43 cases of drug interaction were reported in patients receiving Bisoprolol. The patient demographics comprised predominantly elderly individuals, including multiple patients aged 91 years and a 59-year-old male. Relevant comorbidities included hypersensitivity in one case and cardiac disorders in another, while frequently reported concomitant medications were not specified. The clinical courses exhibited variations but were often characterized by ongoing reports, with all outcomes listed as unknown. The temporal pattern of these interactions was generally not defined, suggesting that the event's occurrence was not systematically reported in correlation with specific drugs or treatment initiation timelines. Based on the available information, a causal relationship between Bisoprolol and drug interaction cannot be established.

Fatigue

During the reporting interval, 33 cases of Fatigue were reported in patients receiving Bisoprolol. The patient population included a diverse group, predominantly consisting of older adults aged 50 and above. Relevant comorbidities included chronic obstructive pulmonary disease and coronary artery disease. Frequently reported concomitant medications included statins and beta-blockers. The clinical course indicated that fatigue was often reported following dosage adjustments and chronic use, with descriptions ranging from mild fatigue to debilitating exhaustion. Reported outcomes varied, with some patients noting improvement with reduction in dosing, while others remained symptomatic. Based on the available information, a causal relationship between Bisoprolol and Fatigue cannot be established.

Safety Section

Safety Section

CCDS / Product Label Overview

The current Company Core Data Sheet (CCDS) / Product Label for Bisoprolol was reviewed as part of this reporting period. Bisoprolol contains bisoprolol fumarate and is classified as a selective beta-1 adrenergic antagonist. The product is available as oral tablets in strengths of 2.5 mg, 5 mg, and 10 mg. Dev Pharma Client holds the marketing authorization for this product.

Indications and Usage

According to the current approved labeling, Bisoprolol is indicated for the management of hypertension and for the treatment of heart failure in patients with reduced ejection fraction. The approved therapeutic use remains consistent with the currently authorized prescribing information.

Contraindications

The current labeling identifies contraindications including severe bradycardia, cardiogenic shock, overt heart failure, and hypersensitivity to bisoprolol or any excipients in the formulation. These contraindications reflect patient populations in whom administration of Bisoprolol may pose unacceptable clinical risk.

Warnings and Precautions

Important warnings and precautions described in the current labeling include a significant risk of cardiovascular events in patients with pre-existing heart disease, particularly heart failure. Patients should be monitored for signs of worsening heart failure and adjustments made accordingly. Abrupt discontinuation of Bisoprolol may lead to exacerbation of angina or myocardial infarction; therefore, a gradual tapering of the dosage is recommended. Additionally, caution is advised in patients with respiratory conditions such as asthma or chronic obstructive pulmonary disease due to the potential for bronchospasm. Particular attention is warranted in diabetic patients, as Bisoprolol may mask symptoms of hypoglycemia. Surgical patients should also be cautioned due to possible variability in hemodynamic responses during surgery.

Drug Interactions

The current labeling describes clinically relevant interactions involving other beta-blockers, calcium channel blockers (specifically non-dihydropyridine types), antiarrhythmics such as amiodarone, and centrally acting agents like clonidine. Furthermore, catecholamine-depleting agents may enhance the hypotensive effects of Bisoprolol. Healthcare providers should consider these interactions when prescribing concomitant therapies to avoid significant alterations in therapeutic efficacy or safety.

Use in Special Populations

Special population considerations include information related to elderly patients, pediatric patients, pregnant women, lactating women, and specific renal or hepatic impairment cases. Current labeling indicates that dose adjustments may be necessary in elderly patients due to potential altered pharmacokinetics. There are insufficient data to support the safe use of Bisoprolol in pediatric populations. It is recommended that pregnant women should only use the medication if the potential benefits justify the risks. Lactating women are advised to consider the benefits of breastfeeding alongside the treatment risks. Caution is also warranted in patients with renal impairment, necessitating careful monitoring and potential dosage adjustments. Hepatic impairment may similarly influence biotransformation and clearance of the drug.

Adverse Reactions

The adverse reaction profile described in the current labeling includes common adverse reactions such as fatigue, dizziness, and bradycardia. Additional postmarketing adverse reactions include hypotension and bronchospasm, which are consistent with the established safety profile of the product. Adverse reactions are generally mild to moderate in severity and are typically dose-dependent.

Laboratory Findings

Reported laboratory abnormalities include increases in liver enzymes and changes in blood glucose levels. These findings are described in the current labeling and should be interpreted within the context of the patient's overall clinical condition. Regular monitoring is recommended for patients receiving long-term treatment with Bisoprolol to identify any potential laboratory abnormalities early.

Overdose Information

The current labeling describes overdose manifestations including severe bradycardia, hypotension, bronchospasm, and heart failure. Recommended management measures include supportive care, intravenous fluids, and, if necessary, the use of atropine or other appropriate pharmacologic agents to reverse symptomatic bradycardia. Continuous clinical monitoring is essential until the patient stabilizes.

CCDS Assessment and Safety Information Updates

No previous CCDS/Product Label version was available for comparison during the reporting interval. Therefore, an assessment of labeling changes could not be performed. No safety-related labeling updates were identified during the reporting interval.

Summary of ICSRs Case Tabulation

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Summary of 15-Day Alert Reports

The following table summarizes Individual Case Safety Reports (ICSRs) submitted as 15-Day Alert Reports during the reporting interval from December 27, 2024, to December 26, 2025. Cases are categorized according to seriousness and expectedness and are further classified by reporting source, including Solicited (Study), Solicited (Other), and Spontaneous reports. These cases represent expedited reporting categories requiring submission within applicable regulatory reporting timelines.

Category	Solicited (Study)	Solicited (Other)	Spontaneous	Total
Serious, Unlabelled — Non-Fatal	10	0	946	956
Serious, Unlabelled — Fatal	0	0	67	67

The tabulation includes serious unlabelled fatal and serious unlabelled non-fatal cases received during the reporting interval. Totals reflect the aggregate number of cases reported from each source category.

Summary of All ICSR Cases

The following table provides a summary of all ICSRs received during the reporting interval. Cases are categorized according to seriousness and labeling status and are presented by reporting source. Reporting sources include solicited reports originating from clinical studies and other organized data collection systems, as well as spontaneous reports received through routine pharmacovigilance activities.

Category	Solicited (Study)	Solicited (Other)	Spontaneous	Total
Serious, Unlabelled	10	0	1013	1023
Non-Serious, Unlabelled	0	0	1	1

The tabulation includes serious labelled, serious unlabelled, non-serious labelled, and non-serious unlabelled cases. Totals represent the overall number of ICSRs received during the reporting interval for each reporting category and source type.